

```
# Chapter 7. Advanced TextGeneration Techniques and Tools
```

```
# `!wget` downloads the Phi-3 Mini 4K Instruct GGUF** model file from the Hugging Face repository and  
# saves it in the current working directory for local use.
```

```
!wget https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf/resolve/main/Phi-3-mini-4k-instruct-fp16.gguf  
  
--2026-07-21 15:31:43-- https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf/resolve/main/Phi-3-mini-4k-instruct-fp16.gguf  
Resolving huggingface.co (huggingface.co)... 108.138.246.79, 108.138.246.67, 108.138.246.85, ...  
Connecting to huggingface.co (huggingface.co)|108.138.246.79|:443... connected.  
HTTP request sent, awaiting response... 302 Found  
Location: https://us.gcp.cdn.hf.co/xet-bridge-us/662698108f7573e6a6478546/a9cdcf6e9514941ea9e596583b3d3c44dd99359fb7dd57f327  
--2026-07-21 15:31:44-- https://us.gcp.cdn.hf.co/xet-bridge-us/662698108f7573e6a6478546/a9cdcf6e9514941ea9e596583b3d3c44dd99359fb7dd57f327  
Resolving us.gcp.cdn.hf.co (us.gcp.cdn.hf.co)... 34.36.28.89  
Connecting to us.gcp.cdn.hf.co (us.gcp.cdn.hf.co)|34.36.28.89|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 7643295904 (7.1G) [application/octet-stream]  
Saving to: 'Phi-3-mini-4k-instruct-fp16.gguf.2'  
  
Phi-3-mini-4k-instr 100%[=====>] 7.12G 8.80MB/s in 76s  
  
2026-07-21 15:33:00 (95.6 MB/s) - 'Phi-3-mini-4k-instruct-fp16.gguf.2' saved [7643295904/7643295904]
```

```
!mv Phi-3-mini-4k-instruct-fp16.gguf.2 Phi-3-mini-4k-instruct-fp16.gguf
```

```
# Uninstalls the existing versions of the LangChain, LangChain Core, and LangChain  
!pip uninstall -y langchain langchain-core langchain-community
```

```
Found existing installation: langchain 0.0.350  
Uninstalling langchain-0.0.350:  
Successfully uninstalled langchain-0.0.350  
Found existing installation: langchain-core 0.1.23  
Uninstalling langchain-core-0.1.23:  
Successfully uninstalled langchain-core-0.1.23  
Found existing installation: langchain-community 0.0.13  
Uninstalling langchain-community-0.0.13:  
Successfully uninstalled langchain-community-0.0.13
```

```
# Installs the the latest versions of LangChain, LangChain Community, and llama-cpp-python  
!pip install -U langchain langchain-community llama-cpp-python
```

```

Stored in directory: /root/.cache/pip/wheels/4a/10/e7/0eb9b120f1640844f33562a3964c5b18b67de1d66d3f9530e8
Successfully built llama-cpp-python
Installing collected packages: requests, httpx-sse, diskcache, llama-cpp-python, pydantic-settings, langsmith, langchain-c
Attempting uninstall: requests
  Found existing installation: requests 2.32.4
  Uninstalling requests-2.32.4:
    Successfully uninstalled requests-2.32.4
Attempting uninstall: langsmith
  Found existing installation: langsmith 0.0.87
  Uninstalling langsmith-0.0.87:
    Successfully uninstalled langsmith-0.0.87
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.34.2 which is incompatible.
google-cloud-bigquery 3.42.2 requires packaging>=24.2.0, but you have packaging 23.2 which is incompatible.
google-adk 2.4.0 requires tenacity<10,>=9, but you have tenacity 8.5.0 which is incompatible.
bigquery-magics 0.14.0 requires packaging>=24.2.0, but you have packaging 23.2 which is incompatible.
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
Successfully installed diskcache-5.6.3 httpx-sse-0.4.3 langchain-1.3.14 langchain-classic-1.0.8 langchain-community-0.4.2

```

Loading Quantized Models with LangChain

```

#from langchain import LlamaCpp (outdated)
# Make sure the model path is correct for your system!

# importing LlamaCpp class that allows the langchain to run the gguf models using llama.cpp backend
from langchain_community.llms import LlamaCpp

llm = LlamaCpp(
    model_path="Phi-3-mini-4k-instruct-fp16.gguf",
    n_gpu_layers=-1,
    max_tokens=500,
    n_ctx=2048,
    seed=42,
    verbose=False
)

```

```

# invoke sends the prompt to the model and return the generated response
llm.invoke("Hi! My name is Maarten. What is 1 + 1?")
# Unfortunately, we get no output! As we have seen in previous chapters,
# Phi 3 requires a specific prompt template

```

```

''

```

```

# updating/installing the langchain package that allows the langchain to interact with open ai or api 's
!pip install -U langchain-openai

```

```

Collecting langchain-openai
  Downloading langchain_openai-1.3.5-py3-none-any.whl.metadata (3.4 kB)
Requirement already satisfied: langchain-core<2.0.0,>=1.4.9 in /usr/local/lib/python3.12/dist-packages (from langchain-openai)
Requirement already satisfied: openai<3.0.0,>=2.45.0 in /usr/local/lib/python3.12/dist-packages (from langchain-openai) (2.4
Requirement already satisfied: tiktoken<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-openai) (0.
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.
Requirement already satisfied: langchain-protocol<0.0.17 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-co
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-co
Requirement already satisfied: uuid-utils<1.0.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langc
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langc
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langc
Requirement already satisfied: jiter<1,>=0.10.0 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langc
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langchain-ope
Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages (from openai<3.0.0,>=2.45.0->langchain-ope
Requirement already satisfied: regex in /usr/local/lib/python3.12/dist-packages (from tiktoken<1.0.0,>=0.7.0->langchain-ope
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from tiktoken<1.0.0,>=0.7.0->langchain-c
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->openai<3.0.0,>=2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai<3.0.0,>=2.4
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai<3.0.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->c
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->l
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->lar
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.
Requirement already satisfied: websockets>=15.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->l
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->lang
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken<

```

Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken<1.0.0,>=1.0.0) 2.0.7
 Downloading langchain_openai-1.3.5-py3-none-any.whl (121 kB)
 121.6/121.6 kB 4.1 MB/s eta 0:00:00
 Installing collected packages: langchain-openai
 Successfully installed langchain-openai-1.3.5

A Single Link in the Chain: Prompt Template

```
## Import PromptTemplate from LangChain's core prompts module
#from langchain import PromptTemplate
from langchain_core.prompts import PromptTemplate

# Create a prompt template with the "input_prompt" variable
template = """<s><|user|>
{input_prompt}
<|end|>
<|assistant|>"""

# Create a PromptTemplate object

prompt = PromptTemplate(
    template=template,          # "template" tells LangChain which prompt format to use
    input_variables=["input_prompt"] # "input_variables" tells LangChain that the template needs a value for "input_prompt"
)
```

```
# creates a basic LangChain chain by connecting the prompt and llm together.
basic_chain = prompt | llm
```

```
# Use the basic chain to send the input to the prompt template and then to the LLM
basic_chain.invoke(
{
    # Provide the actual question or message for the "input_prompt" variable
    "input_prompt": "Hi! My name is Maarten. What is 1 + 1?",
}

)
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
' The answer to 1 + 1 is 2. This is a basic arithmetic operation where you are adding one unit to another, resulting in two
```

```
# Import PromptTemplate from LangChain Core
# PromptTemplate is used to create a reusable prompt with variables
from langchain_core.prompts import PromptTemplate

# Create a prompt template that creates a business name
# {product} is a placeholder that will be replaced with an actual product name
template = """Create a funny name for a business that sells {product}."""

# Create a PromptTemplate object
name_prompt = PromptTemplate(

    # "template" specifies the prompt format we want to use
    template=template,

    # "input_variables" specifies the variable that needs to be provided
    input_variables=["product"]
)

# Connect the prompt template to the language model
# The "|" pipe operator passes the output of name_prompt to llm
# First, the product name is inserted into the prompt
# Then, the completed prompt is sent to the LLM to generate a response
name_chain = name_prompt | llm

# Run the chain by providing "coffee" as the value for the "product" variable
# "coffee" replaces {product} in the prompt
# The final prompt becomes:
# "Create a funny name for a business that sells coffee."

name_chain.invoke({"product": "coffee"})
```

```
'\n<|assistant|> "Caffeinate-O-Matic: The Robotic Barista for Your Caffeine Needs"'
```

A Chain with Multiple Prompts

```
import langchain
print(langchain.__version__)
```

```
1.3.13
```

```
#langchain-classic is mainly used to support older LangChain functionality and code.
!pip install -U langchain-classic
```

```
Requirement already satisfied: langchain-classic in /usr/local/lib/python3.12/dist-packages (1.0.8)
Requirement already satisfied: langchain-core<2.0.0,>=1.4.4 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (2.0.0)
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.2 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (2.0.0)
Requirement already satisfied: langsmith<1.0.0,>=0.1.17 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (0.1.17)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (2.7.4)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (6.0.1)
Requirement already satisfied: requests<3.0.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (2.31.0)
Requirement already satisfied: sqlalchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-classic) (2.0.31)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.33.0)
Requirement already satisfied: langchain-protocol<0.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.1.0)
Requirement already satisfied: packaging<23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (23.1)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (8.2.3)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (4.7.1)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.12.0)
Requirement already satisfied: anyio<3.5.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.5.0)
Requirement already satisfied: distro<1.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.23.0)
Requirement already satisfied: orjson<3.9.14 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.9.14)
Requirement already satisfied: requests-toolbelt<1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.0.0)
Requirement already satisfied: sniffio<1.1 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.1)
Requirement already satisfied: websockets<15.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (10.4)
Requirement already satisfied: xxhash<3.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.0.0)
Requirement already satisfied: zstandard<0.23.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.23.0)
Requirement already satisfied: annotated-types<0.6.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.6.0)
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (2.46.4)
Requirement already satisfied: typing-inspection<0.4.2 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.2.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.2)
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (2.0.7)
Requirement already satisfied: certifi<2023.5.7 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (2023.5.7)
Requirement already satisfied: greenlet<1 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (3.0.3)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.0.2)
Requirement already satisfied: h11<0.16 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (0.14.0)
Requirement already satisfied: jsonpointer<1.9 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.4.4) (1.9)
```

```
# TITLE
```

```
# from langchain import LLMChain
# LLMChain is used to connect a prompt with a language model (LLM).
from langchain_classic.chains import LLMChain

# Import PromptTemplate to create a reusable prompt with a variable
from langchain_core.prompts import PromptTemplate

# Create a chain for the title of our story
template = """<s><|user|>
Create a title for a story about
{summary}
. Only return the title.
<|end|>
<|assistant|>"""

title_prompt = PromptTemplate(template=template, # "template" specifies the prompt format
                              input_variables=["summary"]) # "input_variables" specifies the variable that needs to be passed

# Create an LLMChain by connecting the language model (llm) with the prompt
title = LLMChain(llm=llm, # specifies the language model that will generate the title
                 prompt=title_prompt, # specifies the prompt template used to generate the title

                 # generated result will be stored with the key "title"
                 output_key="title")

#title.invoke({"summary": "a girl that lost her mother"}) JUST FOR CHECKING

/tmp/ipykernel_6208/3717076526.py:13: LangChainDeprecationWarning: The class `LLMChain` was deprecated in LangChain 0.1.17 and will be removed in a future release.
  title = LLMChain(llm=llm, prompt=title_prompt,
```

```
# CHARACTER DESCRIPTION
```

```
# Create a chain for the character description using the summary and title
template = """<s><|user|>
Describe the main character of a story about
{summary}
with the
title
{title}
. Use only two sentences.<|end|>
<|assistant|>"""
character_prompt = PromptTemplate(
template=template, input_variables=["summary", "title"]
)
character = LLMChain(llm=llm, prompt=character_prompt,
output_key="character")
```

```
# Create a chain for the story using the summary, title, and character description
template = """<s><|user|>
Create a story about
{summary}
with the title
{title}
. The main
character is:
{character}
. Only return the story and it cannot be
longer than one paragraph. <|end|>
<|assistant|>"""
story_prompt = PromptTemplate(
template=template, input_variables=["summary", "title",
"character"]
)
story = LLMChain(llm=llm, prompt=story_prompt,
output_key="story")
```

```
# Combine all three components to create the full chain
llm_chain = title | character | story

llm_chain.invoke("a girl that lost her mother")
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
{'summary': 'a girl that lost her mother',
'title': ' "Echoes of a Mother\'s Love: A Girl\'s Journey Through Loss"',
'character': " The protagonist, Lily, is a resilient and empathetic young girl who possesses an innate ability to find
solace in nature after losing her mother. With every leaf she touches, the echoes of her mother's love resonate deeply
within her heart, guiding her through the bittersweet journey of grief and acceptance.",
'story': " Echoes of a Mother's Love: A Girl's Journey Through Loss follows the story of resilient and empathetic young
girl, Lily. The untimely loss of her mother plunged Lily into an abyss of grief, yet she found solace in nature - every
rustle of leaves whispering her mother's love and care to her heart. As seasons changed, so did Lily; each fallen leaf
brought forth the echoes of her mother's warmth, guiding her on a bittersweet journey through sorrow towards acceptance.
Through blossoming trees, flowing rivers, and star-lit nights, Lily's innate connection to nature nurtured her spirit,
ultimately transforming her pain into strength as she discovered the enduring power of love that transcends time and space.
In every leaf, petal, or twinkling starlight, Lily felt an everlasting bond with her mother - a testament to their eternal
love story that continued beyond mortal life."}
```

```
# Memory: Helping LLMs to Remember Conversations
```

```
# Let's give the LLM our name
basic_chain.invoke({"input_prompt": "Hi! My name is Maarten. What is 1 + 1?"})
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
' Hello Maarten! The answer to 1 + 1 is 2. It's a basic arithmetic operation where you add one unit to another unit, result
```

```
# Ask the LLM what name was mentioned in the previous message
# This checks whether the LLM can remember the name "Maarten"

basic_chain.invoke({"input_prompt": "What is my name?"})
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
warnings.warn(
' I'm unable to determine your name without context as I don't have access to personal data about individuals. If you need
```

```
#Memory: Helping LLMs to Remember Conversations
# Conversation Buffer
```

```
#from langchain import PromptTemplate
from langchain_core.prompts import PromptTemplate

# Create an updated prompt template to include a chat history
template = """<s><|user|>Current conversation:
{chat_history}
{input_prompt}
<|end|>
<|assistant|>"""

# Here, we have two variables:
# 1. "input_prompt" = the user's current question or message
# 2. "chat_history" = the previous conversation that we want the LLM to remember
prompt = PromptTemplate(
    template=template,
    input_variables=["input_prompt", "chat_history"]
)
```

```
!pip install -U "langchain==0.0.350" "langchain-community==0.0.13"
```

```
Requirement already satisfied: langchain==0.0.350 in /usr/local/lib/python3.12/dist-packages (0.0.350)
Requirement already satisfied: langchain-community==0.0.13 in /usr/local/lib/python3.12/dist-packages (0.0.13)
Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (6.0.3)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (2.0.36)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (3.8.6)
Requirement already satisfied: dataclasses-json<0.7,>=0.5.7 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (0.6.7)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (1.33.0)
Requirement already satisfied: langchain-core<0.2,>=0.1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (0.1.42)
Requirement already satisfied: langsmith<0.1.0,>=0.0.63 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (0.0.100)
Requirement already satisfied: numpy<2,>=1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (1.26.4)
Requirement already satisfied: pydantic<3,>=1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (2.13.4)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (2.32.3)
Requirement already satisfied: tenacity<9.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.0.350) (8.5.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (0.2.0)
Requirement already satisfied: typing_extensions>=4.4 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (4.12.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain==0.0.350) (1.18.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7,>=0.5.7->langchain==0.0.350) (3.22.0)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7,>=0.5.7->langchain==0.0.350) (0.9.0)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0,>=1.33->langchain==0.0.350) (2.5.0)
Requirement already satisfied: anyio<5,>=3 in /usr/local/lib/python3.12/dist-packages (from langchain-core<0.2,>=0.1->langchain==0.0.350) (4.6.2)
Requirement already satisfied: packaging<24.0,>=23.2 in /usr/local/lib/python3.12/dist-packages (from langchain-core<0.2,>=0.1->langchain==0.0.350) (24.1)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1->langchain==0.0.350) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1->langchain==0.0.350) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1->langchain==0.0.350) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain==0.0.350) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain==0.0.350) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain==0.0.350) (2.2.3)
Requirement already satisfied: certifi>=2023.5.7 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain==0.0.350) (2025.1.1)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from SQLAlchemy<3,>=1.4->langchain==0.0.350) (3.1.1)
Requirement already satisfied: mypy_extensions>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from typing-inspect<1,>=0.4.0->langchain==0.0.350) (1.0.0)
```

```
# ConversationBufferMemory stores the previous messages in the conversation
from langchain.memory import ConversationBufferMemory
```

```
# Import LLMChain to connect the prompt, LLM, and memory together
from langchain.chains import LLMChain
```

```
# Define the type of memory we will use
memory = ConversationBufferMemory(memory_key="chat_history")
```

```
# Chain the LLM, prompt, and memory together
llm_chain = LLMChain(
    prompt=prompt,
    llm=llm,
    memory=memory
)
```

```
# Run the LLM chain with the user's message
llm_chain.invoke({"input_prompt": "Hi! My name is Maarten. What is 1 + 1?"})
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'Hi! My name is Maarten. What is 1 + 1?',
 'chat_history': '',
 'text': " The answer to 1 + 1 is 2. It's a basic arithmetic operation where you add one unit to another, resulting in two
units altogether."}
```

```
# Does the LLM remember the name we gave it?
llm_chain.invoke({"input_prompt": "What is my name?"})
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'What is my name?',
 'chat_history': "Human: Hi! My name is Maarten. What is 1 + 1?\nAI: The answer to 1 + 1 is 2. It's a basic arithmetic
operation where you add one unit to another, resulting in two units altogether.",
 'text': ' Your name is Maarten.'}
```

```
# Windowed Conversation Buffer
```

```
# This type of memory keeps only a limited number of recent conversations
from langchain.memory import ConversationBufferWindowMemory
```

```
# Retain only the last 2 conversations in memory
memory = ConversationBufferWindowMemory(k=2,
memory_key="chat_history")
```

```
# Chain the LLM, prompt, and memory together
llm_chain = LLMChain(
prompt=prompt,
llm=llm,
memory=memory
)
```

```
# Ask two questions and generate two conversations in its memory
```

```
llm_chain.predict(input_prompt="Hi! My name is Maarten and I am 33 years old. What is 1 + 1?")
```

```
llm_chain.predict(input_prompt="What is 3 + 3?")
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
' Hello Maarten! Nice to meet you too. The answer to 3 + 3 is 6.'
```

```
# Check whether it knows the name we gave it
llm_chain.invoke({"input_prompt": "What is my name?"})
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'What is my name?',
 'chat_history': "Human: Hi! My name is Maarten and I am 33 years old. What is 1 + 1?\nAI: Hello Maarten! It's nice to
meet you. The answer to 1 + 1 is 2.\nHuman: What is 3 + 3?\nAI: Hello Maarten! Nice to meet you too. The answer to 3 + 3
is 6.",
 'text': ' Your name is Maarten.'}
```

```
# Check whether it knows the age we gave it
llm_chain.invoke({"input_prompt": "What is my age?"})
```

```
# The LLM indeed has no access to our age since that was not retained in the chat history.
```

```
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'What is my age?',
 'chat_history': "Human: What is 3 + 3?\nAI: Hello Maarten! Nice to meet you too. The answer to 3 + 3 is 6.\nHuman: What
is my name?\nAI: Your name is Maarten.",
 'text': " As an AI, I don't have access to personal data about individuals unless it has been shared with me in the course
of our conversation. Therefore, I cannot determine your age without that information."}
```

```
# Conversation Summary
```

```
# Create a prompt template that summarizes the previous conversation
# and combines it with the new conversation lines
summary_prompt_template = ""<s><|user|>Summarize the
```



```

conversations and update with the new lines.
Current summary:
{summary}
new lines of conversation:
{new_lines}
New summary:<|end|>
<|assistant|>""

```

```

# "new_lines" contains the latest conversation messages
# "summary" contains the existing summary of the previous conversation

```

```

summary_prompt = PromptTemplate(
    input_variables=["new_lines", "summary"],
    template=summary_prompt_template
)

```

```

# Import ConversationSummaryMemory, which summarizes the conversation
from langchain.memory import ConversationSummaryMemory

```

```

# Define the type of memory we will use
memory = ConversationSummaryMemory(
    llm=llm,
    memory_key="chat_history",
    prompt=summary_prompt
)
# Chain the LLM, prompt, and memory together
llm_chain = LLMChain(
    prompt=prompt,
    llm=llm,
    memory=memory
)

```

```

# Generate a conversation and ask for the name
llm_chain.invoke({"input_prompt": "Hi! My name is Maarten. What is 1 + 1?"})
llm_chain.invoke({"input_prompt": "What is my name?"})

```

```

/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'What is my name?',
 'chat_history': " Hello Maarten! The answer to 1 + 1 is 2. It's a basic arithmetic question, but I'm glad you asked—it's
always good to refresh on simple math fundamentals!",
 'text': " Your name is not explicitly mentioned in the current conversation. However, you referred to yourself as
'Maarten' based on your question about the answer to 1 + 1."}

```

```

# Check whether it has summarized everything thus far
llm_chain.invoke({"input_prompt": "What was the first question I asked?"})

```

```

/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/llama_cpp/llama.py:1307: RuntimeWarning: Detected duplicate leading "<s>" in prompt,
  warnings.warn(
{'input_prompt': 'What was the first question I asked?',
 'chat_history': " Hello Maarten! The answer to 1 + 1 is 2. It's a basic arithmetic question, but I'm glad you asked—
it's always good to refresh on simple math fundamentals. Regarding your name, it was not specifically mentioned earlier in
our conversation, but based on your initial query, we identified that you referred to yourself as 'Maarten.'",
 'text': " The first question you asked was 'Hello Maarten! The answer to 1 + 1 is 2. It's a basic arithmetic question,
but I'm glad you asked—it's always good to refresh on simple math fundamentals.'"}

```

```

# Check what the summary is thus far
memory.load_memory_variables({})

```

```

{'chat_history': " Hello Maarten! You first asked for the answer to 1 + 1, which is 2—a basic arithmetic question.
Refreshing on simple math fundamentals can always be beneficial. Additionally, based on your initial query, we identified
that you referred to yourself as 'Maarten.'"}

```

```

# Agents: Creating a System of LLMs

```

```

# Install compatible versions

```

```

!pip install -U --force-reinstall \
    "langchain==0.2.16" \
    "langchain-core==0.2.41" \

```



```
langchain-core==0.1.21" \
"langchain-community==0.2.16" \
"langchain-openai==0.1.25" \
"duckduckgo-search"
```



```

Collecting langchain==0.2.16
  Using cached langchain-0.2.16-py3-none-any.whl.metadata (7.1 kB)
Collecting langchain-core==0.2.41
  Using cached langchain_core-0.2.41-py3-none-any.whl.metadata (6.2 kB)
Collecting langchain-community==0.2.16
  Using cached langchain_community-0.2.16-py3-none-any.whl.metadata (2.7 kB)
Collecting langchain-openai==0.1.25
  Using cached langchain_openai-0.1.25-py3-none-any.whl.metadata (2.6 kB)
Collecting duckduckgo-search
  Using cached duckduckgo_search-8.1.1-py3-none-any.whl.metadata (16 kB)
Collecting PyYAML>=5.3 (from langchain==0.2.16)
  Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting SQLAlchemy<3,>=1.4 (from langchain==0.2.16)
  Using cached sqlalchemy-2.0.51-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (4.3 kB)
Collecting aiohttp<4.0.0,>=3.8.3 (from langchain==0.2.16)
  Using cached aiohttp-3.14.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (8.3 kB)
Collecting langchain-text-splitters<0.3.0,>=0.2.0 (from langchain==0.2.16)
  Using cached langchain_text_splitters-0.2.4-py3-none-any.whl.metadata (2.3 kB)
Collecting langsmith<0.2.0,>=0.1.17 (from langchain==0.2.16)
  Using cached langsmith-0.1.147-py3-none-any.whl.metadata (14 kB)
Collecting numpy<2.0.0,>=1.26.0 (from langchain==0.2.16)
  Using cached numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
Collecting pydantic<3,>=1 (from langchain==0.2.16)
  Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Collecting requests<3,>=2 (from langchain==0.2.16)
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting tenacity!=8.4.0,<9.0.0,>=8.1.0 (from langchain==0.2.16)
  Using cached tenacity-8.5.0-py3-none-any.whl.metadata (1.2 kB)
Collecting jsonpatch<2.0,>=1.33 (from langchain-core==0.2.41)
  Using cached jsonpatch-1.33-py2.py3-none-any.whl.metadata (3.0 kB)
Collecting packaging<25,>=23.2 (from langchain-core==0.2.41)
  Using cached packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
Collecting typing-extensions>=4.7 (from langchain-core==0.2.41)
  Using cached typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting dataclasses-json<0.7,>=0.5.7 (from langchain-community==0.2.16)
  Using cached dataclasses_json-0.6.7-py3-none-any.whl.metadata (25 kB)
Collecting openai<2.0.0,>=1.40.0 (from langchain-openai==0.1.25)
  Using cached openai-1.109.1-py3-none-any.whl.metadata (29 kB)
Collecting tiktoken<1,>=0.7 (from langchain-openai==0.1.25)
  Using cached tiktoken-0.13.0-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (6.7 kB)
Collecting click>=8.1.8 (from duckduckgo-search)
  Using cached click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting primp>=0.15.0 (from duckduckgo-search)
  Using cached primp-1.3.1-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.7 kB)
Collecting lxml>=5.3.0 (from duckduckgo-search)
  Using cached lxml-6.1.1-cp312-cp312-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata (3.5 kB)
Collecting aiohappyeyeballs>=2.5.0 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached aiohappyeyeballs-2.7.1-py3-none-any.whl.metadata (5.9 kB)
Collecting aiosignal>=1.4.0 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached aiosignal-1.4.0-py3-none-any.whl.metadata (3.7 kB)
Collecting attrs>=17.3.0 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting frozenlist>=1.1.1 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached frozenlist-1.8.0-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (20 kB)
Collecting multidict<7.0,>=4.5 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached multidict-6.7.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (5.1 kB)
Collecting propcache>=0.2.0 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached propcache-0.5.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (16 kB)
Collecting yarl<2.0,>=1.17.0 (from aiohttp<4.0.0,>=3.8.3->langchain==0.2.16)
  Using cached yarl-1.24.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (103 kB)
Collecting marshmallow<4.0.0,>=3.18.0 (from dataclasses-json<0.7,>=0.5.7->langchain-community==0.2.16)
  Using cached marshmallow-3.26.2-py3-none-any.whl.metadata (7.3 kB)
Collecting typing-inspect<1,>=0.4.0 (from dataclasses-json<0.7,>=0.5.7->langchain-community==0.2.16)
  Using cached typing_inspect-0.9.0-py3-none-any.whl.metadata (1.5 kB)
Collecting jsonpointer>=1.9 (from jsonpatch<2.0,>=1.33->langchain-core==0.2.41)
  Using cached jsonpointer-3.1.1-py3-none-any.whl.metadata (2.4 kB)
Collecting httpx<1,>=0.23.0 (from langsmith<0.2.0,>=0.1.17->langchain==0.2.16)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting orjson<4.0.0,>=3.9.14 (from langsmith<0.2.0,>=0.1.17->langchain==0.2.16)
  Using cached orjson-3.11.9-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (41 kB)
Collecting requests-toolbelt<2.0.0,>=1.0.0 (from langsmith<0.2.0,>=0.1.17->langchain==0.2.16)
  Using cached requests_toolbelt-1.0.0-py2.py3-none-any.whl.metadata (14 kB)
Collecting anyio<5,>=3.5.0 (from openai<2.0.0,>=1.40.0->langchain-openai==0.1.25)
  Using cached anyio-4.14.2-py3-none-any.whl.metadata (4.6 kB)
Collecting distro<2,>=1.7.0 (from openai<2.0.0,>=1.40.0->langchain-openai==0.1.25)
  Using cached distro-1.9.0-py3-none-any.whl.metadata (6.8 kB)
Collecting jiter<1,>=0.4.0 (from openai<2.0.0,>=1.40.0->langchain-openai==0.1.25)
  Using cached jiter-0.16.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.2 kB)
Collecting sniffio (from openai<2.0.0,>=1.40.0->langchain-openai==0.1.25)
  Using cached sniffio-1.3.1-py3-none-any.whl.metadata (3.9 kB)
Collecting tqdm>=4 (from openai<2.0.0,>=1.40.0->langchain-openai==0.1.25)
  Using cached tqdm-4.69.0-py3-none-any.whl.metadata (57 kB)
Collecting annotated-types>=0.6.0 (from pydantic<3,>=1->langchain==0.2.16)
  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.4 (from pydantic<3,>=1->langchain==0.2.16)
  Using cached pydantic_core-2.46.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.6 kB)
Collecting typing-inspection>=0.4.2 (from pydantic<3,>=1->langchain==0.2.16)
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting charset-normalizer<4,>=2 (from requests<3,>=2->langchain==0.2.16)
  Using cached charset_normalizer-3.4.9-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.met

```